Returning to dynamic pressures: in general, if two stations are fairly close to one another and the disturbances at the two locations are caused by the same source, it is expected that their auto-correlation functions and auto power spectra are of the same general shapes. A plausible model of the correlation function for any two stations x, y was proposed in Reference 32 (Eq. 5-8, p. 43) as:

$$R_{xy}(r,\tau) = \sigma_x \sigma_y \exp(-\delta |r|) \rho_o (\tau - \tau') \tag{4}$$

where σ_x, σ_y denote the rms values (with zero mean), r is the distance between the two points, δ is the spatial decay coefficient, τ is time, and τ' relates to the time of convection of the pressure from one point to the other. ρ_o denotes the normalized auto-correlation function. Based on the above assumption of identical or near-identical auto-correlation functions (different only in amplitude), the phase angle θ of the cross spectral density ϕ_{xy} and the cross-correlation coefficient as defined by Eq. (3) may be expressed in the following:

$$\theta = -2\pi f \tau' \tag{5}$$

$$\rho_{xy}(r,\tau') \simeq \exp(-\delta|r|) \simeq \gamma_{xy}(r, f) \tag{6}$$

In Eq. (6), γ_{xy} is the square root of the coherence function. Apparently, γ_{xy} is assumed to be independent of frequency f. Based on Eq. (6), the spatial decay constant δ may be determined. Processing of flight test data indicated that the above condition was true only in limited locations for certain flight maneuvers. For instance, based on dynamic pressure data obtained in Run 2, Flight 871 (M_o = 0.925, h = 10,668 m), the contours of equal γ_{xy} were plotted in Figure 19 for Station Number 2. The contours were plotted only in the area where the convection effect was observed and the approximate formulation of Eq. (6) was applicable.

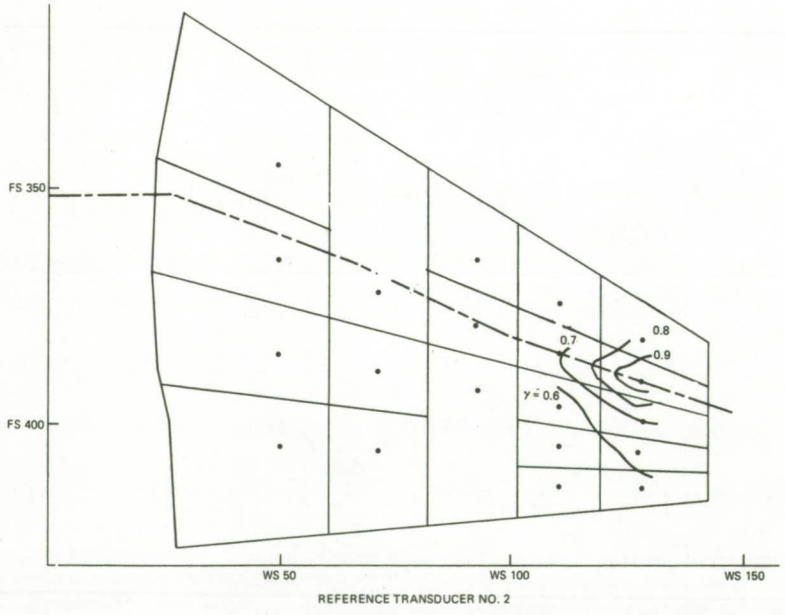


FIGURE 19. CONTOUR OF MEAN SQUARE ROOT VALUES OF COHERENCE FUNCTION γ_{xy} FOR BUFFET PRESSURES OBTAINED IN RUN 2, FLIGHT 871, M_o = 0.925, h = 10,668 m, $\delta_n = 4°$, $\delta_f = 12°$